

# Control work 7 — speed and motion

The formula, its rearrangements, average speed and two-body problems.

LESSON 132

1 × 40 MIN

MATEMATIKA 6, NAZORAT ISHI 7

S7 12 REVIEW

LESSON CLOCK

2'

28'

8'

2'

|                       |        |
|-----------------------|--------|
| Instructions          | 2 min  |
| The paper             | 28 min |
| Answers and diagnosis | 8 min  |
| What comes next       | 2 min  |

LEARNING OBJECTIVES

- Use  $s = vt$  in all three directions with correct units.
- Find an average speed over a journey in stages.
- Solve a meeting or overtaking problem.
- Classify each lost mark and rewrite the whole solution.

TEXTBOOK

|           |                      |
|-----------|----------------------|
| National  | pp. 335–388          |
| Cambridge | Stage 7, pp. 122–126 |
| Workbook  | Control paper 7      |

01

## Explanation

### The paper — 20 marks, 28 minutes

| Q | TASK                                                             | MARKS | FROM     |
|---|------------------------------------------------------------------|-------|----------|
| 1 | A car does 75 km/h for 4 h: the distance                         | 2     | L117–119 |
| 2 | 4 km in 20 min: the speed in km/h                                | 3     | L120–122 |
| 3 | 7 km at 5 km/h: the time in hours and minutes                    | 3     | L120–122 |
| 4 | Convert 72 km/h to m/s                                           | 2     | L116     |
| 5 | 120 km at 60 and back at 40: the average speed                   | 4     | L126–128 |
| 6 | Two cars 300 km apart at 60 and 40: when and where do they meet? | 4     | L129–131 |
| 7 | A bus at 60 leaves 30 min before a car at 80: when is it caught? | 2     | L129–131 |

THE ANSWERS

300 km; 12 km/h; 1 h 24 min; 20 m/s; 48 km/h; after 3 h, 180 km from the first; after 1.5 h.

Naming the slip

| SLIP                                     | WHAT IT LOOKS LIKE         | THE FIX                    |
|------------------------------------------|----------------------------|----------------------------|
| minutes used with km/h                   | $4 \div 20$                | $4 \div \frac{1}{3}$       |
| decimal hours read as minutes            | $1\text{ h }40\text{ min}$ | $1\text{ h }24\text{ min}$ |
| multiplied instead of divided            | $72 \cdot 3.6$             | $72 \div 3.6$              |
| speeds averaged                          | $(60 + 40) \div 2$         | $240 \div 5$               |
| closing speed subtracted for an approach | $60 - 40$                  | $60 + 40$                  |
| head start left as a time                | $0.5 \div 20$              | $30 \div 20$               |
| units missing                            | $300$                      | $300\text{ km}$            |

Name the slip in the margin, then rewrite the whole solution — not the wrong line.

What comes next

| IF YOU LOST MARKS ON | REVISE                                    |
|----------------------|-------------------------------------------|
| Q1–Q3                | the formula and time conversion, L116–122 |
| Q4                   | unit conversion, L116                     |
| Q5                   | average speed, L126–128                   |
| Q6–Q7                | two-body problems, L129–131               |

LOOKING FORWARD

The rest of Quarter III is a Cambridge project, two lessons on fractions and sequences, and then the volume of the cuboid — where a formula again does the work.

## 02 Worked examples

**EXAMPLE 1** Model answer, Q2: 4 km in 20 minutes.

$$① \quad 20 \text{ min} = \frac{1}{3} \text{ h.}$$

Convert first.

$$② \quad v = 4 \div \frac{1}{3} = 4 \cdot 3$$

$$③ \quad = 12 \text{ km/h.}$$

Reasonable for a cyclist ✓

ANSWER

12 km/h

**EXAMPLE 2** Model answer, Q5: 120 km at 60 and back at 40.

$$① \quad \text{Out: 2 h. Back: 3 h.}$$

$$② \quad \text{Total 240 km in 5 h.}$$

$$③ \quad 48 \text{ km/h.}$$

Not the mean of the speeds.

ANSWER

48 km/h

**EXAMPLE 3** Model answer, Q6: two cars 300 km apart.

- ① Closing speed 100 km/h.  
They approach, so add.

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- ② Time 3 h.

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- ③ First travels 180 km, second 120 km.  
Check: 300 ✓

**ANSWER**

After 3 h, 180 km from the first

**03** Interactive model

Return Q5 first and ask how many wrote 50; naming the error as “averaging the speeds” makes it memorable.

**The chapter in eight questions**

75 km/h for 4 h:

18.75

300

79

300 km/h

4 km in 20 min:

0.2

12

80

4

7 km at 5 km/h:

1.4 h

1 h 40 min

1 h 24 min

35 h

72 km/h in m/s:

20

259

26

7.2

120 km at 60 and back at 40:

50

48

100

240

Two cars approaching close the gap at:

the sum

the difference

the average

the larger

300 km apart at 60 and 40: they meet after:

3 h

5 h

7.5 h

2 h

A 30 min head start at 60 is:

0.5 km

30 km

60 km

120 km

## 04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

| ENGLISH       | O'ZBEKCHA            | РУССКИЙ            |
|---------------|----------------------|--------------------|
| Control work  | Nazorat ishi         | Контрольная работа |
| Speed         | Tezlik               | Скорость           |
| Distance      | Masofa               | Расстояние         |
| Time          | Vaqt                 | Время              |
| Average speed | O'rtacha tezlik      | Средняя скорость   |
| Closing speed | Yaqinlashish tezligi | Скорость сближения |
| Units         | O'lchov birligi      | Единицы            |
| Diagnosis     | Tashxis              | Диагностика        |

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

**05 Quick check****Check yourself**

Q2 needs the time in:

minutes

hours

seconds

any unit

Q3 answer in hours and minutes is:

1.4 h

1 h 40 min

1 h 24 min

84 min only

Q4 divides by:

3.6

1000

60

36

Q5 must not:

add the distances

average the speeds

add the times

divide once

Q6 adds the speeds because:

they travel together

they approach

they are equal

of the units

Q7 turns the head start into:

a time

a distance

a speed

nothing

## 06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. 75 km/h for 4 h

ANSWER 300 km

2. 4 km in 20 min

ANSWER 12 km/h

3. 7 km at 5 km/h

ANSWER 1 h 24 min

4. 72 km/h in m/s

ANSWER 20

5. 240 km in 5 h

ANSWER 48 km/h

6. Closing speed of 60 and 40 approaching

ANSWER 100 km/h

7. A 30 min head start at 60

ANSWER 30 km

Medium · the standard question

7 PROBLEMS

1. 120 km at 60 and back at 40

ANSWER 48 km/h

2. Two cars 300 km apart at 60 and 40

ANSWER Meet after 3 h

3. Where?

ANSWER 180 km from the first

4. A bus at 60 caught by a car at 80 after a 30 min start

ANSWER 1.5 h

5. 54 km/h in m/s

ANSWER 15

6. 9 km at 4 km/h

ANSWER 2 h 15 min

7. 150 km in 2 h 30 min

ANSWER 60 km/h

**Hard · stretch and reason**

7 PROBLEMS

1. A journey of 200 km, half at 40 and half at 60

ANSWER 48 km/h

2. Two trains 540 km apart at 100 and 80

ANSWER Meet after 3 h

3. A car catches a lorry 45 km ahead in 1.5 h: the speed difference

ANSWER 30 km/h

4. A cyclist at 15 km/h for 40 min then 20 km/h for 30 min: the average

ANSWER 17.1 km/h

5. A train 180 m long passing a pole at 15 m/s

ANSWER 12 s

6. Why is the average of 60 and 40 not 50 here?

ANSWER More time is spent at 40

7. 12 m/s in km/h

ANSWER 43.2

## 07 Homework

### Homework — 5 tasks

Rewrite every question you lost a mark on, with its units.

1. Rewrite in full every question on which you lost a mark.
2. A car covers 294 km in 3 h 30 min. Find its speed.
3. Find the average speed for 60 km at 30 km/h and 60 km at 60.
4. Two cyclists 45 km apart ride towards each other at 12 and 18 km/h. When do they meet?
5. Convert 90 km/h to m/s and 25 m/s to km/h.